

Problem

The inductors in Fig. 6.87 are initially charged and are connected to the black box at $t = 0$. If $i_1(0) = 4 \text{ A}$, $i_2(0) = -2 \text{ A}$, and $v(t) = 50e^{-200t} \text{ mV}$, $t \geq 0$, find:

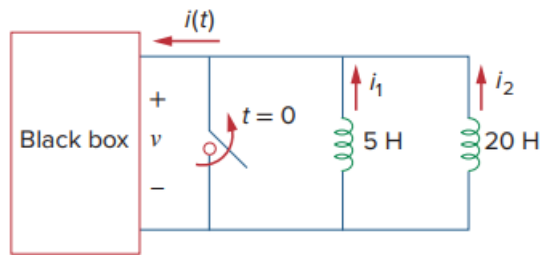
(a) the energy initially stored in each inductor,

(b) the total energy delivered to the black box from $t = 0$ to $t = \infty$,

(c) $i_1(t)$ and $i_2(t)$, $t \geq 0$,

(d) $i(t)$, $t \geq 0$.

Fig. 6.87:



Step-by-step solution

Step 1 of 6

Given data

$$i_1(0) = 4 \text{ A},$$

$$i_2(0) = -2 \text{ A},$$

$$v(t) = 50e^{-200t} \text{ mV for } t \geq 0$$

Step 2 of 6

(a)

Initial energy stored in inductor 5H is

$$W_{5H} = \frac{1}{2} L_1 [i_1(0)]^2$$

$$W_{5H} = \frac{1}{2} (5) (4)^2$$

$$\boxed{W_{5H} = 40 \text{ J}}$$

Initial energy stored in inductor 20H is

$$W_{20H} = \frac{1}{2} L_2 [i_2(0)]^2$$

$$W_{20H} = \frac{1}{2} (20) (-2)^2$$

$$\boxed{W_{20H} = 40 \text{ J}}$$

Step 3 of 6

(b)

Total energy delivered to the black box from $t = 0$ to $t = \infty$ is

$$W_{\text{Total}} = W_{5H} + W_{20H}$$

$$W_{\text{Total}} = 40J + 40J$$

$$\boxed{W_{\text{Total}} = 80J}$$

Step 4 of 6

(c)

Current flowing through inductor $5H$ is

$$i_1(t) = \frac{1}{L_1} \int_0^t -v(t) dt + i_1(0)$$

$$i_1(t) = \frac{1}{5} \int_0^t -(50e^{-200t} \times 10^{-3}) dt + 4$$

$$i_1(t) = \frac{1}{5} \times \frac{-50 \times 10^{-3}}{-200} e^{-200t} \Big|_0^t + 4$$

$$i_1(t) = 5 \times 10^{-5} (e^{-200t} - 1) + 4$$

$$\boxed{i_1(t) = [5 \times 10^{-5} (e^{-200t} - 1) + 4] \text{ A}} \quad \text{for } t > 0$$

Step 5 of 6

Current flowing through inductor $20H$

$$i_2(t) = \frac{1}{L_2} \int_0^t -v(t) dt + i_2(0)$$

$$i_2(t) = \frac{1}{20} \int_0^t -50e^{-200t} \times 10^{-3} dt - 2$$

$$i_2(t) = \frac{1}{20} \times \frac{-50 \times 10^{-3}}{-200} e^{-200t} \Big|_0^t - 2$$

$$i_2(t) = 1.25 \times 10^{-5} (e^{-200t} - 1) - 2$$

$$\boxed{i_2(t) = [1.25 \times 10^{-5} (e^{-200t} - 1) - 2] \text{ A}} \quad \text{for } t > 0$$

Step 6 of 6

(d)

Total current flowing into black box for $t \geq 0$ is

$$i(t) = i_1(t) + i_2(t)$$

$$i(t) = 5 \times 10^{-5} (e^{-200t} - 1) + 4 + 1.25 \times 10^{-5} (e^{-200t} - 1) - 2$$

$$\boxed{i(t) = [6.25 \times 10^{-5} (e^{-200t} - 1) + 2] \text{ A}} \quad \text{for } t > 0$$